A Course End Project Report on

Smart Parking System using Arduino

A8603 –Internet of Things

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This is to certify that the Course End Project titled **"Smart Parking using Arduino"** is carried out by **" M.Varun Tej Reddy (24881A0530)"**, "**M.Rohan (24881A0531)"**, " **M.Yashwanth (24881A0532)"**, and **"Md.Ibrahim (24881A0533)"**, towards **A8603–Internet of Things Laboratory** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** during the Academic Year **2025–2026.**

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Abstract

The Smart Parking System using Arduino is an innovative solution designed to address the growing problem of vehicle parking in urban areas. With increasing population and vehicle usage, finding available parking spaces has become a major challenge, leading to traffic congestion, fuel wastage, and time loss. This project aims to provide an automated and efficient parking management system using Arduino and sensors.

The system uses IR sensors to detect the presence of vehicles in parking slots. The Arduino processes the sensor data and updates the availability status of each slot. LEDs are used to indicate the status of parking spaces (green for available, red for occupied). Additionally helping drivers quickly identify free spaces. The system can be extended with IoT integration to allow remote monitoring via mobile applications.

Keywords: Arduino Uno, Smart Parking, IR Sensor, Servo motor, IoT, Automation.

1.Introduction

Parking management is a critical issue in modern cities.
Traditional parking systems rely on manual monitoring, which is inefficient and prone to errors. Drivers often waste time searching for available spaces, contributing to traffic congestion and pollution.

To overcome these issues, a Smart Parking System using Arduino is developed. This system automatically detects the occupancy of parking slots using sensors and provides real-time updates to users. It reduces human effort, saves time, and improves resource utilization.

2. Literature Review

Smart parking systems have been widely researched to improve urban mobility. Early systems used manual ticketing and attendants, which were inefficient. With advancements in embedded systems, microcontroller-based solutions have become popular.

Researchers have implemented parking systems using infrared and ultrasonic sensors for vehicle detection. Recent developments include IoT-enabled parking systems that allow users to check availability remotely via mobile apps. These systems improve efficiency, reduce congestion, and promote sustainable urban living.

3.Objectives

The main objectives of the Smart Parking using Arduino are:

- To design a system that automatically detects parking slot occupancy.
- To provide real-time indication of available and occupied slots.
- To reduce traffic congestion and time spent searching for parking.
- To develop a cost-effective and reliable solution.
- To enable future integration with IoT for remote monitoring.

4.Methodology

4.1 Architectural Diagram

The Smart Parking System is implemented using Arduino Uno, IR sensors, LEDs, servo motor

- IR sensors are placed in each parking slot to detect vehicle presence.
- Arduino Uno processes sensor data and updates slot status.
- LEDs indicate slot availability (green = free, red = occupied).
- The system operates continuously, providing real-time updates.

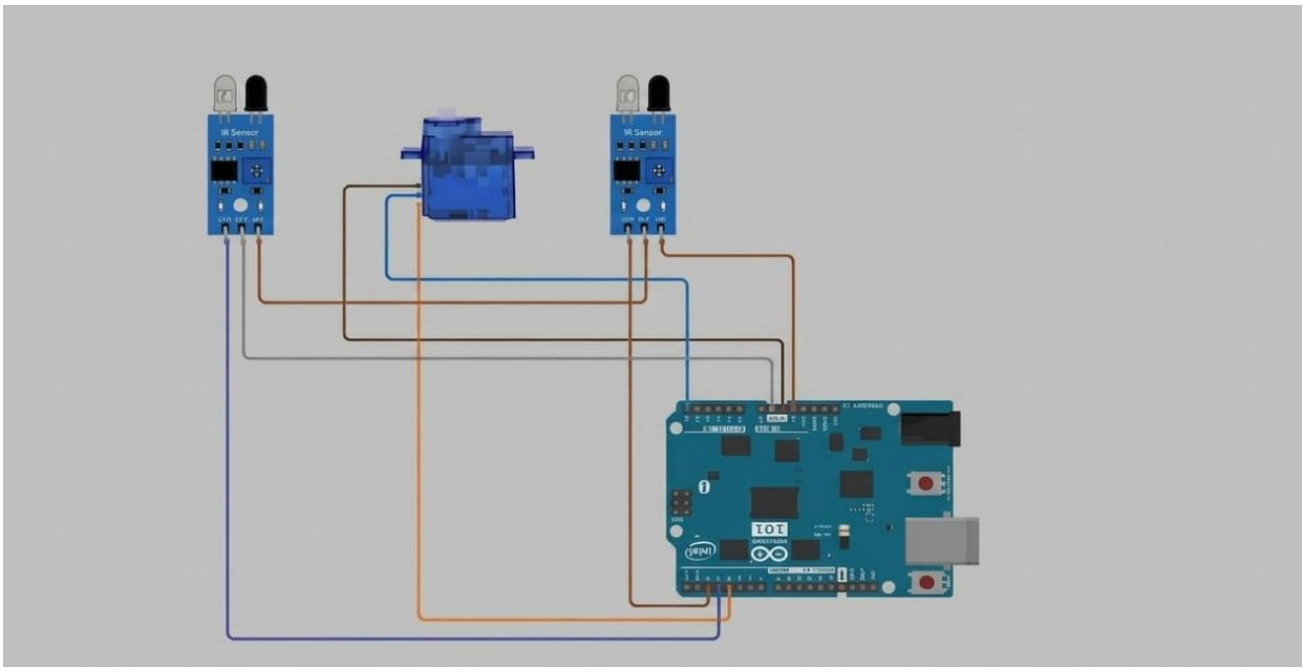

Figure 1:Circuit Diagram of Smart Parking System using Arduino

4.2 Hardware and Software Requirements

Hardware: The following hardware components are used to implement the Smart Parking System using Arduino:

- Arduino Uno: Acts as the main controller that processes the sensor data and controls all components.
- **IR Sensors (HC-SR04):** Detect the presence of vehicles by measuring distance in each parking slot.
- **Servo motor: A servo motor is a small motor that rotates to a specific angle (like 0°–180°) using control signals, commonly used for precise movement in projects like robots and Arduino.**
- **LEDs (Red, Green):** Indicate the status of parking slots, with red for occupied and green for available.
- **Jumper Wires:** Used for making electrical connections.
- **USB Cable / Power Supply:** Provides power to the Arduino.

Software: The following software is used to develop and run the system:

- **Arduino IDE:** Used to write, compile, and upload the program to the Arduino board.
- **Embedded C (Arduino Programming Language):** Used to write the logic for reading sensor data and controlling outputs.

4.3 Working Principle

The working principle of the Smart Parking System using Arduino revolves around the use of ultrasonic sensors to detect the presence or absence of vehicles in individual parking slots. Each parking slot is equipped with an IR sensor, which continuously emits high-frequency sound waves and measures the time taken for the waves to reflect back after hitting an object, such as a parked vehicle.

When a vehicle occupies a parking slot, the ultrasonic sensor detects a shorter distance due to the proximity of the vehicle surface. This measured distance is compared against a predefined threshold value that distinguishes between an occupied and an empty slot. If the distance measured is less than the threshold, the system interprets the slot as occupied; otherwise, it is considered available.

The Arduino microcontroller acts as the central processing unit of the system. It receives distance measurements from the IR sensors and processes this data to determine the status of each parking slot. Based on the occupancy status, the Arduino controls the output devices such as LEDs

LED indicators are used to provide a visual representation of slot availability: a green LED lights up to indicate a free slot, while a red LED signals that the slot is occupied. Simultaneously, allowing drivers to quickly identify free spaces without the need to physically check each slot.

4.4 Software Implementation

```
#include <Servo.h>

Servo gate;

// IR pins
const int irEntry = 2;
const int irExit  = 3;

// Parking count
int carCount = 0;
const int maxSlots = 4;

// State tracking (to avoid multiple triggers)
int lastEntryState = HIGH;
int lastExitState  = HIGH;

void setup() {
  pinMode(irEntry, INPUT);
  pinMode(irExit, INPUT);

  gate.attach(9);
  gate.write(0); // Gate closed

  Serial.begin(9600);
  Serial.println("Smart Parking System Started");
}

void loop() {

  int entryState = digitalRead(irEntry);
  int exitState  = digitalRead(irExit);

  //     ENTRY
  if (entryState == LOW && lastEntryState == HIGH) {

    if (carCount < maxSlots) {
      Serial.println("Car Entering...");

      gate.write(90);   // Open gate
      delay(2500);

      gate.write(0);    // Close gate
      delay(1000);

      carCount++;
      Serial.print("Cars inside: ");
      Serial.println(carCount);
    }
    else {
      Serial.println("Parking FULL");
```

```
    }
  }

  //  EXIT
  if (exitState == LOW && lastExitState == HIGH) {

    Serial.println("Car Exiting...");

    gate.write(90);
    delay(2500);

    gate.write(0);
    delay(1000);

    if (carCount > 0) {
      carCount--;
    }

    Serial.print("Cars inside: ");
    Serial.println(carCount);
  }

  // Save states
  lastEntryState = entryState;
  lastExitState  = exitState;

  delay(100);
}
```

5. Results and Discussion

The Smart Parking System was thoroughly tested across multiple parking slots under various conditions to evaluate its performance and reliability. The IR sensors consistently detected the presence or absence of vehicles with high accuracy, effectively distinguishing between occupied and vacant slots based on distance measurements.

The Arduino microcontroller processed the sensor inputs in real-time, promptly updating the status indicators. The LEDs provided clear and immediate visual feedback, with red LEDs signaling occupied slots and green LEDs indicating availability. The complemented this by showing the total number of free parking spaces, enabling users to quickly assess parking availability without physically inspecting each slot.

During testing, the system demonstrated robustness and efficiency in managing parking space information, significantly reducing the time drivers spent searching for available spots. However, minor fluctuations in sensor readings were observed, which could be attributed to environmental factors such as weather conditions, sensor alignment, or the presence of small objects within the detection range.

Overall, the system's performance met the design objectives, proving to be a reliable and cost-effective solution for smart parking management. The real-time updates and intuitive indicators contribute to smoother traffic flow and enhanced user convenience.

Future improvements could focus on integrating wireless communication for remote monitoring and expanding the system to accommodate larger parking facilities.

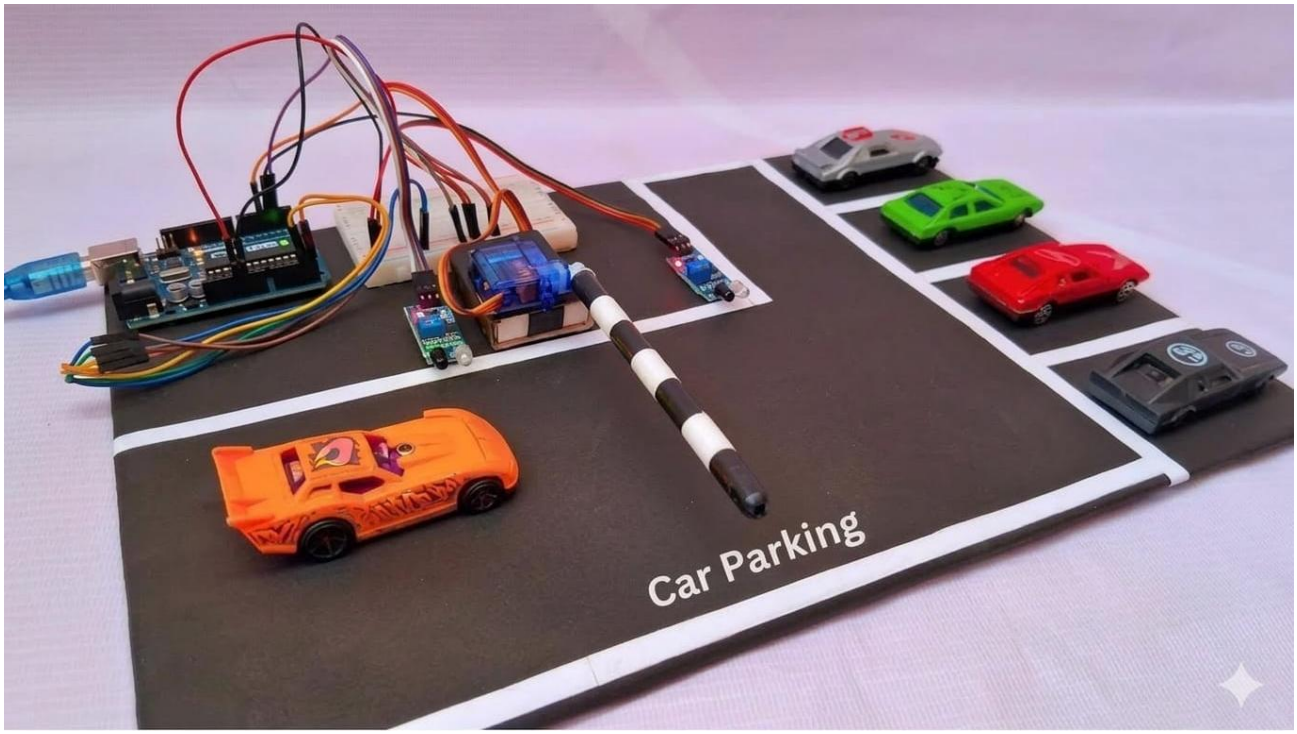

Figure 2:Experimental Setup of Smart Parking System using Arduino

6. Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied embedded system concepts in design and implementation.
PO2	Problem Analysis	Identified limitations of manual parking systems.
PO3	Design/Development of Solutions	Developed a functional smart parking application.
PO5	Engineering Tool Usage	Used Arduino IDE and sensors for development.
PO6	Engineer and Society	Helps society by reducing congestion and pollution.

6.2 Sustainable Development Goals (SDGs) Mapping

SDG No.	Goal	Relevance
SDG 4	Quality Education	Demonstrates practical application of IoT concepts.
SDG 9	Industry, Innovation, and Infrastructure	Promotes smart city infrastructure.
SDG 11	Sustainable Cities and Communities	Improves urban mobility and reduces congestion.
SDG 12	Responsible Consumption and Production	Promotes efficient use of resources.

7. Conclusion

The Smart Parking System using Arduino is a cost-effective and efficient solution for managing parking spaces. It successfully detects slot occupancy and provides realtime updates, reducing congestion and saving time.

This project highlights the practical application of embedded systems and sensor technologies in addressing urban challenges. The integration of IR sensors with Arduino provides a reliable and automated approach to parking management,enhancing user convenience and optimizing space utilization.

Future enhancements could focus on incorporating IoT capabilities for remote monitoring, mobile app integration for user interaction, and scalability to accommodate larger parking facilities. These improvements would further increase the system's effectiveness and adaptability in smart city environments, making it a valuable tool for modern urban infrastructure.

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